

Introduction to IOT Systems

Learning Outcomes



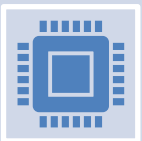
Understand the definition and significance of the Internet of Things,



Discuss the architecture, operation, and business benefits of an IoT solution,

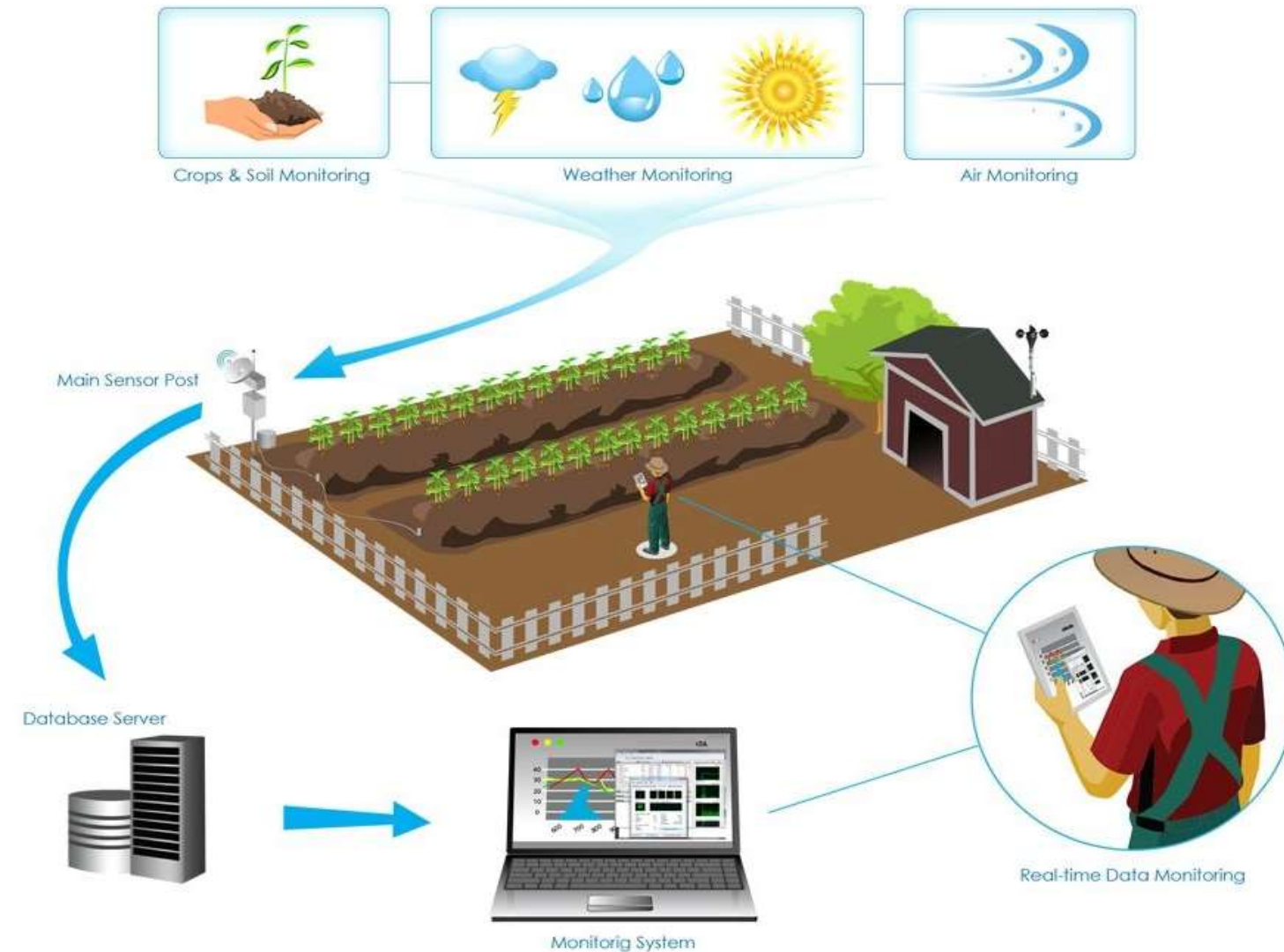


Identify different microcontrollers and available features,



Develop simple applications using to Microcontroller based Development Systems.

What is IoT?



- The **Internet of Things (IoT)** describes physical objects - “**things**”- that are embedded with sensors, software, and other technologies, and that connect and exchange data with other devices and systems over the **Internet**.

Importances of IOT Technology

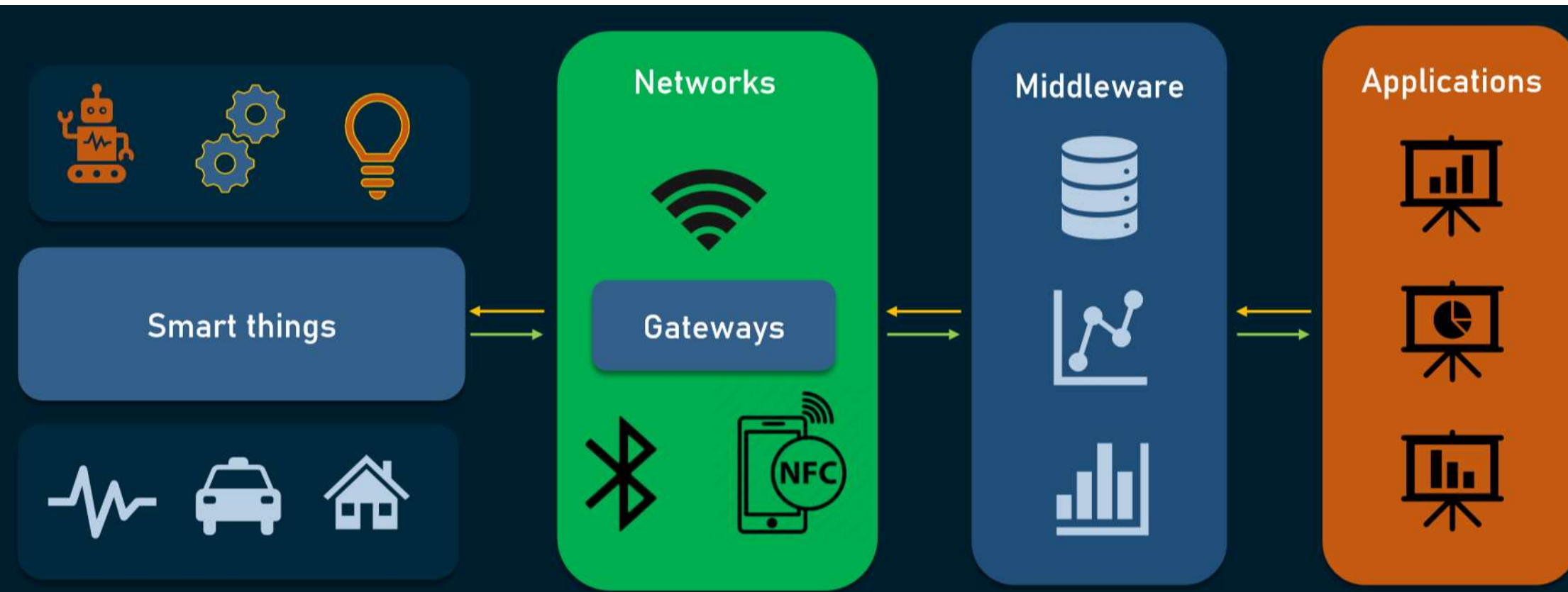
IOT has 6 Any architecture

- **Any-thing** - Manufacturing, Automotive, home appliances, vehicles, All Industries
- **Any-one** - Human to Human (H2H), Human to Thing (H2T), Thing to Thing (T2T)
- **Any-time** - IoT can used for 24hrs x 365 days
- **Any-place** - Outdoors, Indoors, Under water or Under Ground
- **Any-service** – Healthcare, Education, Public Sector
- **Any-network** - Wi-Fi, GSM, GPRS, Ethernet

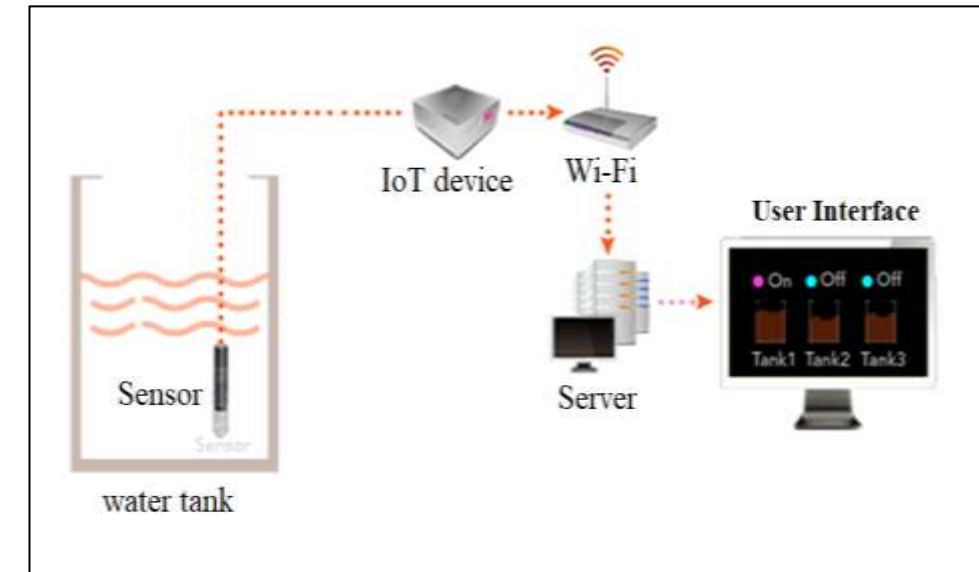
Can you Measure the temperature at your home, school and father's office at same time?

Main Components of IoT System

- Main components in a complete IoT system:
 - **Sensors/ devices**
 - **Network/ Connectivity**
 - **Data processing**
 - **User interface**



The IoT Lifecycle



- **Collect data** - Sensors are used to collect data from environment
- **Communicate data** - Sending data through network to the destination
- **Analyze Data** – Analyze the data and produce information
- **Action on Data**- Taking decision / action based on the analyzed data

Advantages of IoT Technology

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- Ability to access information from **anywhere** at any time on any device.
 - Objects can **connect to the internet** via embedded devices.
 - **All-in-one communication** is possible between people, processes, and things.
 - **Minimize human efforts** in many life aspects.
 - **Reduce the cost** of production (saving time and money).
 - Make decisions **faster and accurately**.
 - Use **low-power** sensor technology.
 - **Secured** data processing.



IoT Applications

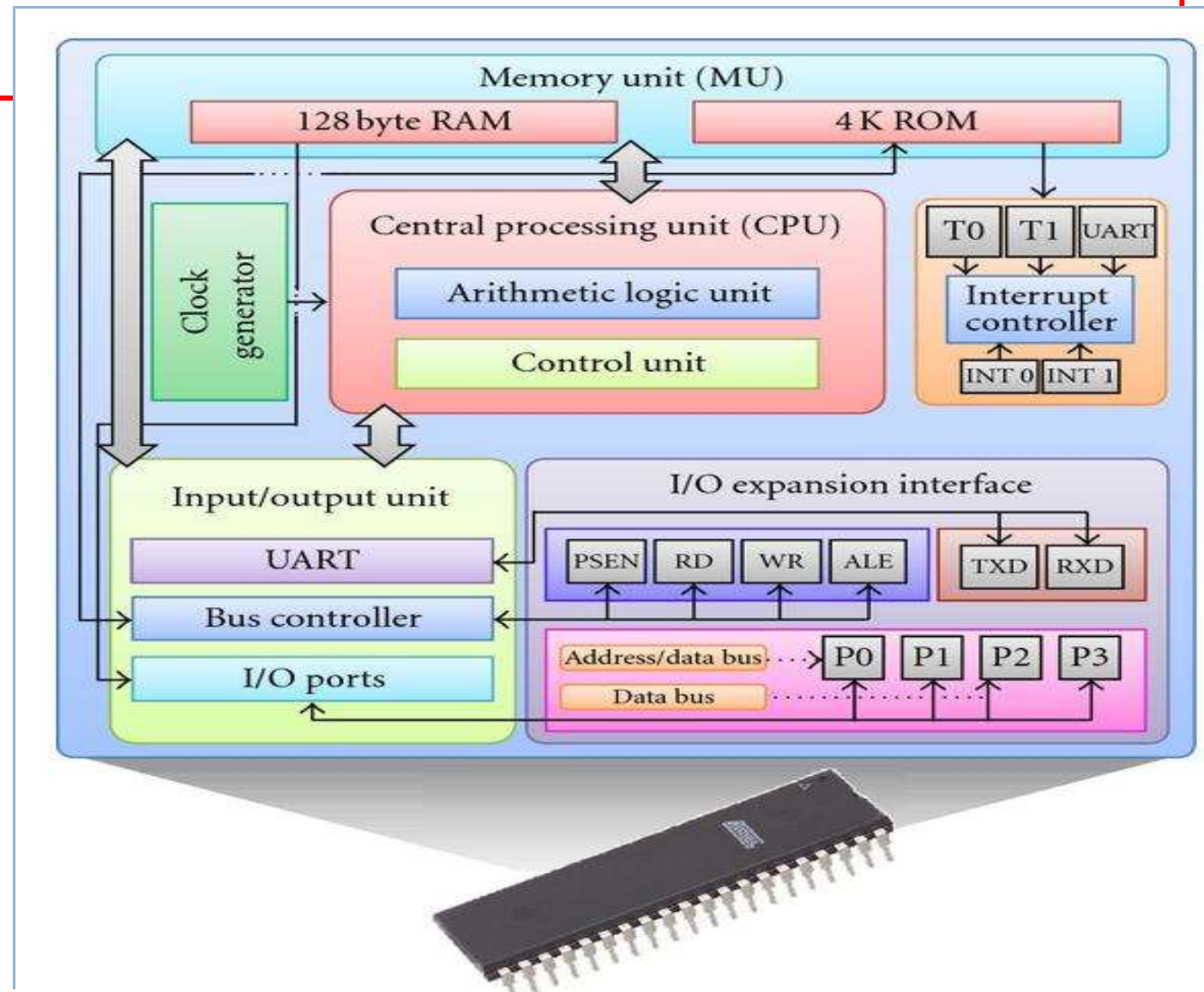
- Smart Homes
- Smart City
- Self-driven Cars
- IoT Retail Shops
- Health Care
- Agriculture
- Safety and Security

What is a Microcontroller ?

A microcontroller is a **small computer** present in a **single integrated circuit** which is dedicated to perform **one task** and execute **one specific application**.

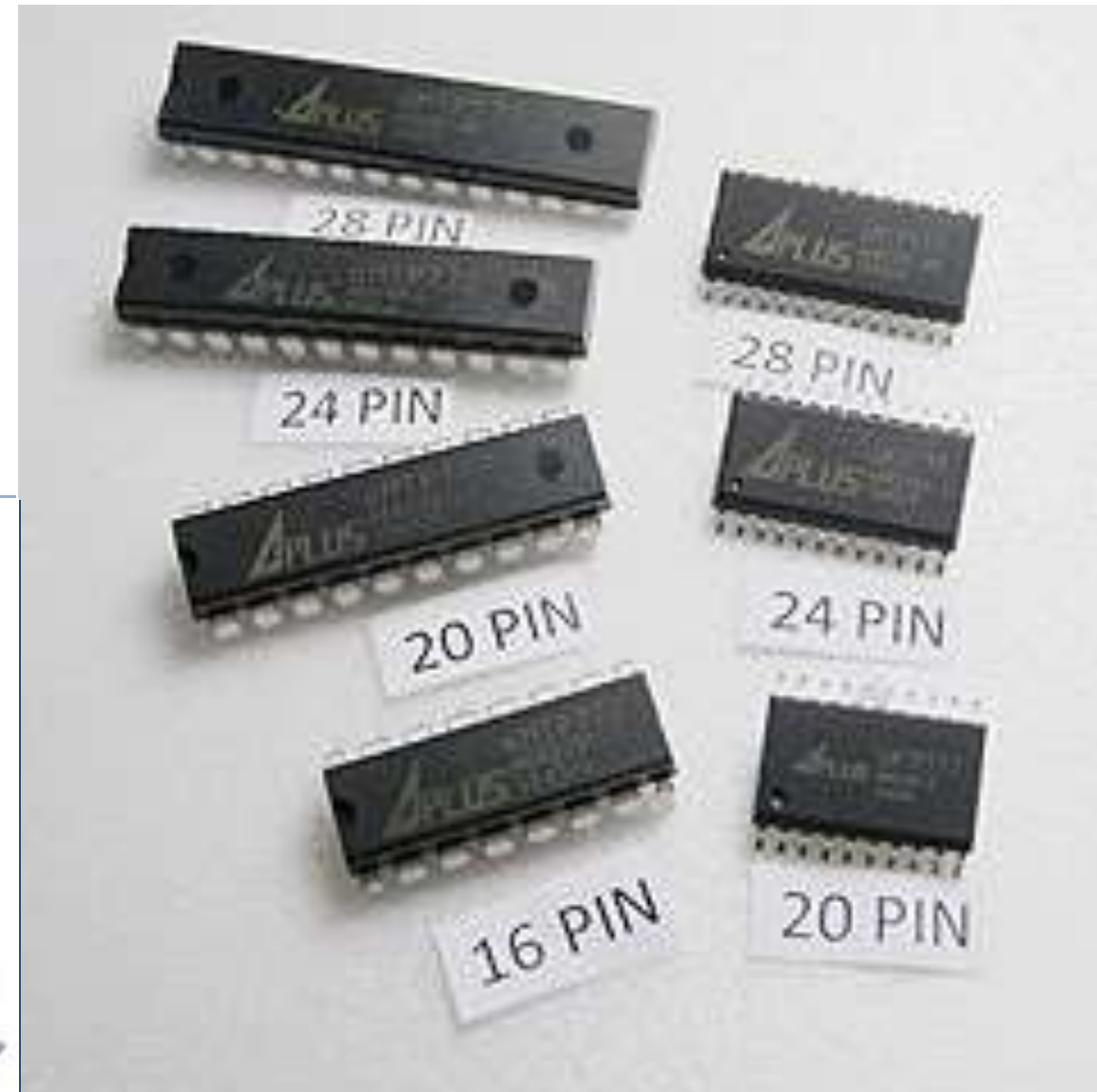
Microcontroller contains CPU with memory and programmable input/output peripherals.

- Microprocessor- CPU unit
- Memory unit
- I/O pins unit
- Clock unit



List of Microcontroller family

- **PIC** microcontroller
- **Atmel** microcontroller
- **Intel** microcontroller
- **EPSON** microcontroller

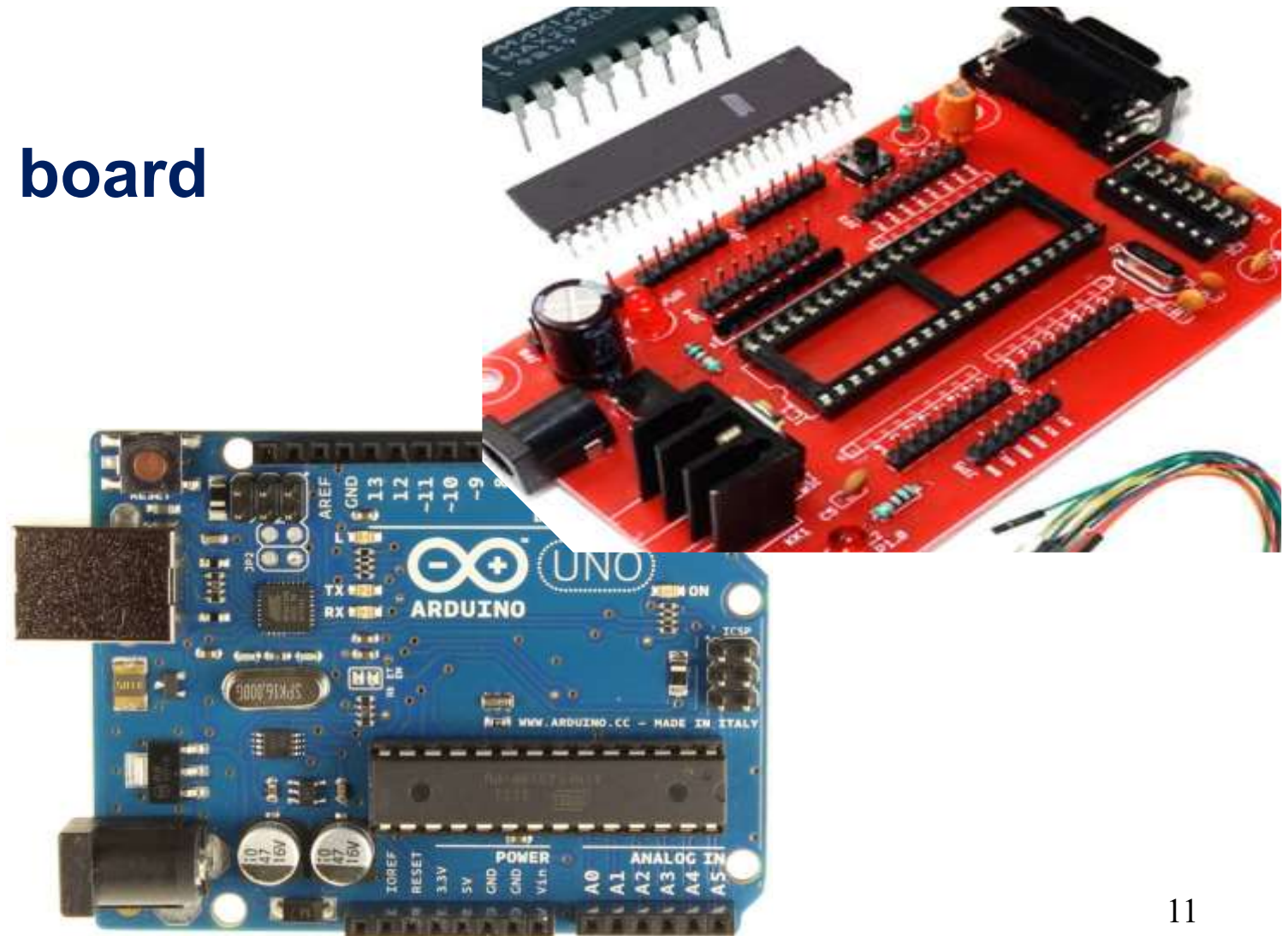


Microcontroller Development Boards

- A microcontroller development board is a circuit board that can be used to **programme** and **test** with a specific microcontroller boards capabilities.

Peripherals in development board

- Digital I/O pins
- Analog I/O pins
- Power supply unit/ pins
- IC Bases
- Programming interface
- USB ports
- Communication ports



Microcontroller Programming Software

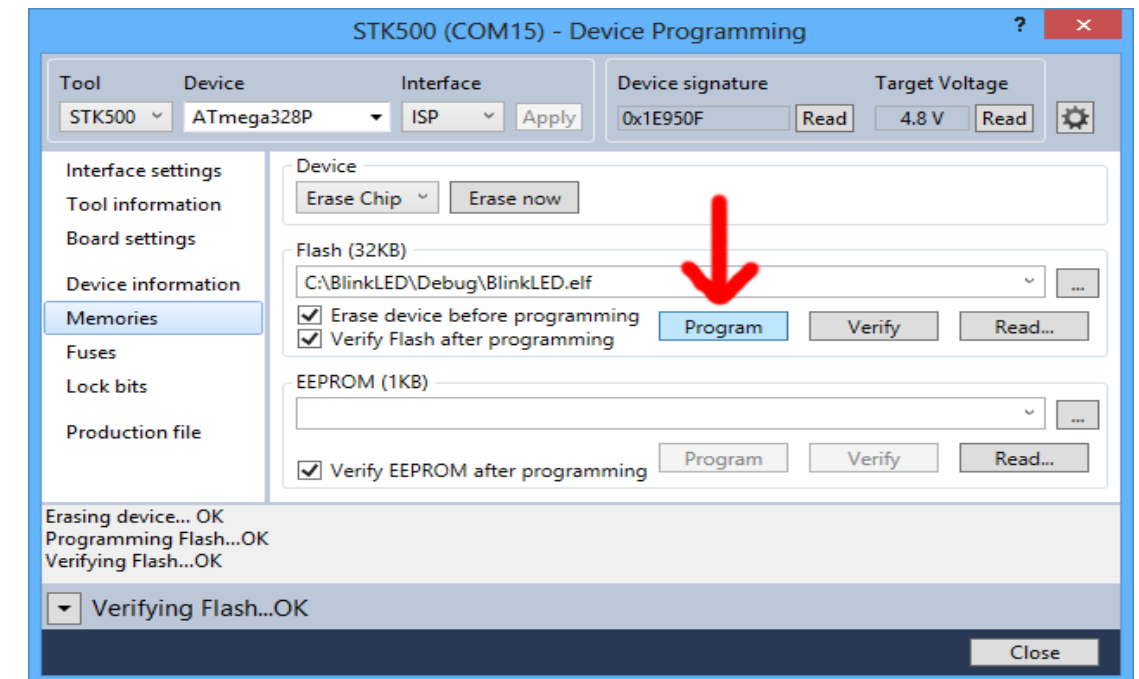
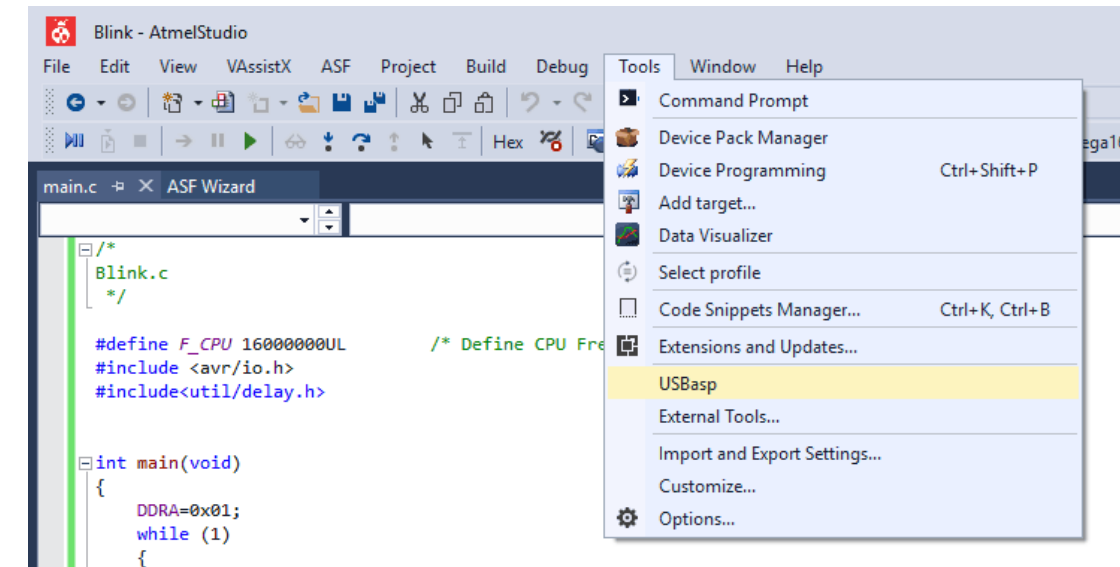
PIC Microcontroller:

- MICROCHIP MPLAB® IDE
- HI-TECH C Compiler
- PIC Programmer
- PIC Debugger



ATMEL Microcontroller:

- Atmel Studio
- AVR-GCC
- AVR Studio



Input and Output Devices for Microcontrollers

